

Overwind

Winding Engineer's Assistant • Kerrow No. 3, twelve days before the licence

12 DAYS • 40 STOPS • GRADE 12 • OVERWIND

THE OPENING CARD

Everyone at Kerrow rides down twelve hundred and forty metres of shaft in a cage, hung on one steel rope. Last March it went past its landing with the brake put on at the right mark, and the pads took the blame. You are the winding engineer's assistant, and the winding profile is yours to hand over. The inspector reads it in twelve working days, and a plan to cut twelve seconds off each trip turns on it. Forty-one men a shift ride what he signs.

What the profile says

BEFORE THE DAY — TRIAL

Walk the yard before the inspection

Five buildings, a headframe and a conveyor, and the rope crosses the yard at head height between two of them.

PLAN CARD 178W

Monday. The inspector comes in twelve working days to re-license the winder. Delia Marchetti, the winding engineer, wants twelve seconds out of every cycle. What she has handed in is a curve. It shows how fast the cage should be going at each second of a trip. A curve is not a set of numbers yet. What you hand over is the profile the inspector signs: how fast the cage may run, and how hard it may be stopped. A number with no working behind it does not go in. Grace Ubah, the banksman, meets every one of those trips at the landing. She reads the same curve as a shape, not as a plan. Today you get the acceleration out of Marchetti's curve. Acceleration is how fast the speed itself changes. From it you can work out what the cage floor has to push up with, and where her twelve seconds really come from. Wilf Teague, a development miner, has ridden the six o'clock wind for nine years and has never seen the sheet that decides it.

BRIEFING 18W

Twelve working days to the inspector, and the proposal on the desk is a shape nobody has differentiated.

WHAT YOU DECIDE 15W

Get the acceleration and the depth out of the shape the winder is driven to.

1.1 Twelve seconds of ramp

THE BANK · A ROOM · DERIVE

SCENE 45W

Ubah has the new ramp on the landing desk. The winder is to accelerate at 1.2 metres a second squared at the start. The acceleration then tapers to nothing over twelve seconds. She wants the speed at the end of it, and the rope wound.

GUIDE 55W

You build the working a line at a time, and each line follows the one above. The ramp is stated as an acceleration. Speed comes from accumulating it, and distance from accumulating the speed. Both start from zero here, which fixes the constants. The taper is what makes this an integral rather than a multiplication.

ASK 18W

What speed does the cage reach at the end of the ramp, and how much rope is wound?

BEHIND THE BUTTON 152W

Why the ramp tapers. An acceleration that switches on and off in a step is felt as a jolt. Everybody in the cage feels it. What a passenger notices is how fast the acceleration is changing. Tapering it to nothing at the end of the ramp is what makes a fast wind bearable.

Why the constant-acceleration formulas will not do. They are derived on the assumption that the acceleration does not change, and here it changes throughout. Using them with the peak value overstates the speed by a factor of two. Using the mean gets the speed right and the distance wrong.

What the two answers are for. The speed at the end of the ramp is what the rest of the cycle is driven from. The distance is what says whether the ramp fits. There are 1,240 metres of shaft, and the crawl at the top needs its share of them.

TAKES AS READ

an acceleration is the rate at which a velocity changes

VERDICT 83W

The cage reaches 7.2 metres a second, having wound 57.6 metres. The ramp is stated as an acceleration that falls linearly to nothing. The speed at any instant is the accumulated acceleration up to then. The distance is the accumulated speed. Each integration brings a constant. Both are fixed by the cage starting from rest at the landing. Using the peak acceleration for the whole twelve seconds would give 14.4 metres a second. That is twice what the winder is being asked for.

TAKEAWAY 13W

Integrating an acceleration gives a velocity, and integrating that gives a distance. Each

1.2 What the floor has to push with

SHAFT AND BRAKE HOUSE · A PERSON · BALLPARK

SCENE 41W

Aziz Rahimi, the shaft engineer, has the cage on the bench drawing. Four tonnes of cage and men, accelerating upward at the peak of the ramp. He wants the force the floor has to push up with while that is happening.

GUIDE 58W

Five quantities are offered and three of them belong in the sum. The body being drawn round is the cage and everybody in it. Two forces act on it: the floor pushing up and gravity pulling down. The rope's own mass is on the list because it is not part of this body, and neither is the drum.

ASK 15W

What force does the cage floor push up with at the peak of the ramp?

TAKES AS READ

a net force upward means the upward forces exceed the downward ones

VERDICT 91W

The floor pushes with about 44 kilonewtons. Drawing the body is the whole of the work. Once the cage and the men are one object, only two forces act on it. The second law fixes their difference. The floor therefore pushes with more than the weight while the cage is speeding up. That is what everybody in it feels. On the way into the stop it pushes with less than the weight. Anything outside the drawn body — the rope, the drum, the headframe — belongs to a different free-body diagram.

TAKEAWAY 23W

The force a floor pushes with is the weight plus whatever the acceleration needs, and only the chosen body's mass belongs in it.

1.3 Where twelve seconds come from

WINDER HOUSE · A ROOM · CHOICE

SCENE 30W

Marchetti has the two cycles side by side on the winder house desk, the present one and the one she wants. The full-speed section is the same length in both.

GUIDE 54W

All four options are places a cycle could be shortened. Ask of each what it would change about the curve on the desk. Then ask which parts of that curve the depth of the shaft has already fixed. Only one of the four has any seconds in it that are not already spoken for.

ASK 9W

Where do Marchetti's twelve seconds have to come from?

TAKES AS READ

a journey is made of a ramp up, a run at speed, a ramp down and a slow approach

VERDICT 86W

They come out of the ramps and the crawl. The full-speed section is fixed by the depth of the shaft and the speed the winder may run at. Shortening it means winding faster, not accelerating harder. The loading time at each landing is fixed by how long men take to get on and off. What is left is how hard the cage is accelerated, and how long it spends creeping into the landing. That is why this proposal is a proposal about acceleration and about margin.

TAKEAWAY 19W

A cycle time is the sum of parts fixed by different things. Only some of them are anybody's to

END OF THE DAY 15W

A speed curve contains the acceleration and the distance, and neither is written on it.

What the drum is

PLAN CARD 130W

Tuesday. The submission gives a moment of inertia for the drum with no working behind it. Moment of inertia is how hard a spinning thing is to speed up or slow down. Marchetti took the figure from a handover file that is eighteen years old. The drum has been re-flanged since then. Samuel Otieno, the chief rope examiner, has read the same file. It has the rope forty metres longer than the rope really is. Today you work the drum's inertia out from the drum's own size. You also have to settle what the rope's eleven tonnes hang from, and go down to the inset to see Tom Halloway, the onsetter. Every torque in the submission rests on that one number. So does every stopping sum, and every cycle time.

BRIEFING 15W

The torque calculation on the desk uses a figure for the drum nobody can source.

WHAT YOU DECIDE 16W

Get the drum's inertia out of how its mass is arranged, not out of its weight.

2.1 Eighteen tonnes, and where it sits

WINDER HOUSE · A ROOM · DERIVE

SCENE 38W

The drum is a steel cylinder of eighteen tonnes, 2.1 metres to its outside and 1.9 to its bore, turning about its axis. Marchetti wants its moment of inertia from the drum itself rather than from the file.

GUIDE 65W

You build the working a line at a time. The quantity wanted is a sum over the whole body. Each piece contributes its mass times the square of its distance from the axis. So the first decision is what a piece is. Choose it so that every part of the piece is the same distance out. The integral then becomes one in a single variable.

ASK 11W

What is the drum's moment of inertia about its own axis?

BEHIND THE BUTTON 166W

Why a thin shell is the right element. Mass at different radii contributes differently, so an element has to be all at one radius. A thin cylindrical shell of radius r and thickness dr is exactly that. Its mass is the total mass times its share of the cross-sectional area.

Why the fourth powers collapse. Integrating r^3 gives $r^4/4$. So the shells leave a difference of fourth powers over a difference of squares. That factors, because a difference of squares is what a difference of fourth powers is made of. What survives is the mean of the two squared radii, which is why the answer looks so simple.

Why this is not the mass. A solid cylinder of the same eighteen tonnes and the same outside radius has an inertia of 39,700 kilogram metres squared. This one comes out at 72,200. The solid drum's mass is spread inward towards the axis, where it counts for little. The two machines behave nothing alike and weigh the same.

TAKES AS READ

the moment of inertia of a small mass is that mass times the square of its distance from the axis · angular velocity and angular acceleration — taken as read

VERDICT 87W

The drum's inertia is about 72,200 kilogram metres squared. Taking a thin shell as the element makes the integral possible. Every part of a shell is the same distance from the axis. The shell's mass is the total mass times its share of the cross-section, so the integral is in r alone. Integrating r^3 between the two radii leaves a difference of fourth powers. That factors into the mean of the squares. A solid drum of the same mass would come out at about half of this.

TAKEAWAY 21W

A moment of inertia is an integral over the body. Two bodies of one mass can differ in it by a

2.2 Eleven tonnes, hung from what

ROPE SHOP · A PERSON · CHOICE

SCENE 34W

Otieno has the rope's own weight written on the shop board: about eleven tonnes with the cage at the inset. The submission's force calculation has it appearing in the cage's line and nowhere else.

GUIDE 54W

Four accounts of where the rope's weight is carried. Ask of each which body has been drawn round. A force belongs to a diagram rather than to a site. The distinction that decides it is between what hangs below a point and what is held above it. The answer changes as the cage moves.

ASK 10W

Whose free-body diagram does the hanging rope's weight belong in?

TAKES AS READ

a force appears in the equation of whichever body it acts on

VERDICT 83W

It belongs to the drum, through the tension at the top of the rope. Each point of the rope carries everything below it. So the tension grows from the cappel at the bottom to the drum at the top. The eleven tonnes is fully accounted for only at the drum. The cage carries nothing of it: the rope pulls up on the cage, not down. This is why the tension in the submission has to be a function of where the cage is.

TAKEAWAY 19W

A force belongs to whichever body it acts on. Choosing the body is the first move in every problem

2.3 Three things out of one trace

THE BANK · A ROOM · PROTOCOL

SCENE 25W

Halloway has the depth recorder's paper from the March wind: one pen, one trace, position against time, and nothing else recorded anywhere on the plant.

GUIDE 51W

Each thing the inquiry wanted on the left has one operation on the right. That operation would get it from a trace of position against time. What separates them is which direction you have to go. Differentiating gives rates. Accumulating gives totals. One of the four is on the trace already.

ASK 15W

Match each thing the inquiry wanted to what has to be done to the trace.

TAKES AS READ

a trace of position against time contains every rate derived from it

VERDICT 84W

A position record contains everything derived from it. Speed is its slope. Retardation is the slope of the speed. How fast the retardation is changing is one slope further down. Three quantities, out of one pen. The distance past the landing is on the paper directly, because it is a difference of positions. Accumulating is the other way, and you need it when the record is a rate rather than a position. That is the case at the winder, where the tachometer reads speed.

TAKEAWAY 19W

A single record of position against time yields speed, acceleration and jerk by differentiating, and needs no other instrument.

END OF THE DAY 21W

How hard something is to spin up depends on where its mass sits, not on how much of it there is.

The rope is heavier than the cage

PLAN CARD 136W

Wednesday. Otieno has stopped the submission going forward. The tension in it is the cage and its load times gravity. That is the pull at the cappel, up at the top of the cage. It is not the pull anywhere a rope ever breaks. With the cage at the inset there are twelve hundred and forty metres of steel hanging above it. The rope weighs nearly three times what the cage carries. Today you find the tension at the drum with the rope's own weight in it. Then you work out what a man standing in the cage feels as the stop begins. Ruth Anand, the winder driver, will tell you how a hard ramp differs from a soft one. Marchetti's whole case rests on a margin worked out at the wrong end of the rope.

BRIEFING 17W

The submission's tension figure is one number, and the rope's weight changes with where the cage is.

WHAT YOU DECIDE 19W

Get the tension at the drum with the rope's own weight in it, and see what a body feels.

3.1 Where the tension is worst

ROPE SHOP · A ROOM · DERIVE

SCENE 35W

Otieno has the rope's mass per metre chalked on the shop board and the cage at the inset on the depth indicator. Marchetti's ramp accelerates it upward at 1.2 metres a second squared from there.

GUIDE 69W

You build the working a line at a time. Draw round the cage and everything hanging with it. Then ask what has to be true of the pull at the top of that body. Two things go into it. One is the weight of what is hanging, which depends on how much rope is out. The other is the extra the acceleration needs. Note which way the acceleration points.

ASK 15W

What is the tension at the drum with the cage accelerating upward from the inset?

BEHIND THE BUTTON 160W

Why the worst case is at the bottom of the shaft. The rope carries its own weight, so the tension at any point is set by everything below that point. With the cage at the inset, everything below the drum is the whole rope plus the cage. With the cage at the bank it is almost nothing.

Why the acceleration multiplies rather than adds. The body has to be given a net upward force, and every kilogram of it needs its share. So the acceleration acts on the rope's eleven tonnes as well as on the cage's four. That is where two thirds of the extra tension comes from.

What the number is for. The rope's breaking load is about 1,600 kilonewtons. The statutory factor of safety at this depth is eight, which puts the working limit at 200. The figure in the submission is 39 kilonewtons. Both of those are true statements about different points on the same rope.

TAKES AS READ

a rope's tension at a point carries everything hanging below that point

VERDICT 87W

The tension is about 166 kilonewtons, against the 39 in the submission. Drawing round the cage and the rope together makes the tension at the drum the only unknown force on that body. The second law then fixes it: the hanging mass times gravity, plus the same mass times the acceleration. The rope contributes eleven of the fifteen tonnes. Accelerating upward increases the tension, and decelerating reduces it. That is why the worst instant of a wind is the start of an upward one from the bottom.

TAKEAWAY 20W

Tension at the drum is the hanging mass times gravity plus the acceleration, and the hanging mass includes the rope.

3.2 Eighty-two kilograms, going up and slowing

SHAFT AND BRAKE HOUSE · A PERSON · BALLPARK

SCENE 39W

Rahimi wants the figure that goes in the men's notice. Somebody of eighty-two kilograms is standing on the floor. The cage is rising and being retarded at 1.1 metres a second squared. What does he feel through the floor?

GUIDE 60W

Five quantities are offered and three of them make the answer. The body is the man, not the cage, so the cage's four tonnes is on the list to be left out. The retardation is the one to take, because the cage is going up and slowing down, and the sign of that is what the whole question turns on.

ASK 17W

What does the floor push a man of eighty-two kilograms with, on the way into the stop?

TAKES AS READ

the floor's push is the only upward force on somebody standing on it

VERDICT 76W

The floor pushes with about 714 newtons, against 804 while standing still: a man feels about eleven per cent lighter. Rising and slowing down means the acceleration points downward, so the floor has less to do than hold him up. The mistake worth avoiding is the one about bodies rather than the one about signs. The cage's four tonnes is carried by the rope. It appears nowhere in the forces on a person standing inside it.

TAKEAWAY 17W

Going up and slowing down feels lighter, because the floor is pushing with less than the weight.

3.3 What the driver feels

WINDER HOUSE · A ROOM · CHOICE

SCENE 37W

Anand has driven this winder for twelve years. She says a hard ramp and a soft one of the same length feel like different machines. The difference is not in how fast the cage ends up going.

GUIDE 49W

All four options are quantities that change when a ramp is made harder. Ask of each whether it is what a passenger standing in a cage would actually notice. Then ask which of them can be different in two ramps that reach the same speed in the same time.

ASK 16W

What is different between a hard ramp and a soft one that reach the same speed?

TAKES AS READ

an acceleration can itself change while a body is moving

VERDICT 84W

It is how quickly the acceleration itself changes. Two ramps can reach the same speed over the same twelve seconds and get there differently. One switches to full acceleration at once and holds it. The other eases in and out. What a body feels is the force through the floor. A sudden change in acceleration therefore arrives as a jolt. That rate of change is a quantity in its own right: the slope of the acceleration, one derivative below what the submission's curve shows.

TAKEAWAY 19W

A rate can have a rate of its own. A body standing on a floor feels the second one.

3.4 Twelve seconds of ramp — Review

THE BANK · A ROOM · CALLBACK · BALLPARK

SCENE 40W

Grace Ubah, the banksman, has the other end of the same journey on the landing desk. The retardation ramp starts at 1.5 metres a second squared and tapers to nothing over eight seconds. She wants the speed it takes off.

GUIDE 61W

Five quantities are offered, and three of them belong in the sum. The ramp is stated as a rate that falls linearly to nothing. What it removes is the area under it, not a product of its peak and its length. Two of the five belong to the other end of the journey, or to the speed the ramp starts from.

ASK 9W

How much speed does the retardation ramp take off?

TAKES AS READ

an acceleration that changes cannot be multiplied by a time

VERDICT 81W

The ramp takes off six metres a second. That leaves the cage creeping into the landing at 1.2. A retardation stated as a rate that falls linearly to nothing removes the area under itself, not its peak times its length. The mean of a linear taper is half its peak. So eight seconds at a peak of 1.5 do the work of eight at 0.75. Using the peak throughout would take twelve off a wind that has only 7.2 to give.

TAKEAWAY 19W

A rate that tapers to nothing does half the work its quoted peak suggests. What it delivers is the

END OF THE DAY 22W

The tension in a hoisting rope is a function of where the cage is, and its worst value is at the drum.

The torque is there

BEFORE THE DAY — TRIAL

The bench track, and the hut at the end of it

The gravity station is three hundred metres out along the bench and the track is peat for the last hundred. Take it once before you need to, because the day Dr. Lena Sokol, the geophysicist, has the shaft for two hours you will be walking it with an instrument that must not be jarred.

PLAN CARD 141W

Thursday, and this is the day Marchetti has waited a fortnight for. The drum's inertia has been measured off the drum. The tension has been taken at the right end of the rope. So the torque the faster wind asks for can now be worked out instead of claimed. Torque is the twist the motor has to supply. Otieno has said all week that he is not arguing about torque. Iona Craig, the brake fitter, has a different question. She wants to know what the pads can hold once the motor has been given those numbers. Today you take the torque at the hardest instant of the cycle. You also settle what a friction limit is, and put the drum and the rope into one relation with Ubah. If the torque comes in under the motor's rating, the submission goes forward.

BRIEFING 15W

Marchetti's case comes down to one number in newton metres, and today it gets computed.

WHAT YOU DECIDE 14W

Find the torque the faster wind needs, and say whether the machine has it.

4.1 What the motor is asked for

WINDER HOUSE · A ROOM · DERIVE

SCENE 36W

The drum is 2.1 metres to the rope. Its inertia is 72,200 kilogram metres squared. The tension at the start of the upward wind is 166 kilonewtons. Marchetti wants the torque the motor has to produce.

GUIDE 61W

You build the working a line at a time. The drum is the body. Two torques act on it, the motor's and the rope's. Their difference is what spins it up. The angular acceleration is not given. It has to come from the cage's acceleration and the radius the rope leaves at. Watch which of the two torques opposes the other.

ASK 15W

What torque does the motor have to produce at the start of an upward wind?

BEHIND THE BUTTON 178W

Why the load torque is the larger part. The rope pulls on the drum at 2.1 metres from the axis with 166 kilonewtons. That is 348 kilonewton metres before the drum is asked to change speed at all. The inertia term adds 41 on top of it. That is a tenth of the total, and the part that only exists while the ramp is running.

Why the two are tied. The rope does not slip on the drum. So the cage's speed is the drum's angular speed times the radius. The two accelerations are tied the same way. That is what turns one unknown into the other. It is the only place the radius does two different jobs in one line.

What the answer settles and what it does not. The motor is rated at 480 kilonewton metres continuous, so 389 is inside it with room to spare. Marchetti is right about the machine. Nothing in this calculation contains a period, a stretch or a rope that talks back, which is the argument that arrives on day nine.

TAKES AS READ

a rope that does not slip ties the cage's acceleration to the drum's angular acceleration

VERDICT 83W

The motor has to produce about 389 kilonewton metres. Two torques act on the drum. The angular form of the second law says their difference equals the inertia times the angular acceleration. So the motor's torque is the rope's plus that product. The angular acceleration comes from the rope not slipping: the cage's 1.2 metres a second squared over the 2.1 metre radius. The load term dominates. That is why a bigger drum makes the problem worse rather than better on both counts.

TAKEAWAY 20W

The torque a wind needs is the load torque plus the inertia torque. The second is the smaller of the

4.2 A number measured cold

SHAFT AND BRAKE HOUSE · A PERSON · CHOICE

SCENE 32W

Craig has her own certificate from the inquiry on the bench. Four things are written on it: a coefficient, an instrument, a pressure and a temperature. The finding quoted the first one.

GUIDE 50W

Four readings of what a friction coefficient tells you about a brake. Ask of each what the measurement actually did. A coefficient sets the most a surface can hold, not what it is holding at any moment. The distinction that decides this card is between a limit and a value.

ASK 8W

What does Craig's coefficient tell the winder house?

TAKES AS READ

a friction force takes whatever value is needed up to a limit set by the surfaces

VERDICT 89W

It gives the most the pads can hold, at the pressure and temperature she measured. Friction is not a fixed force: below the limit it supplies exactly what is needed and nothing more, and at the limit it slips. So a coefficient is a ceiling. It is a ceiling under stated conditions. A hot drum at the end of a wind is not a cold plate on a bench. The inquiry quoted the number without the conditions, which turns a careful measurement into a claim it was never making.

TAKEAWAY 20W

A friction coefficient gives the largest force a contact can hold, and it carries the conditions it was measured under.

4.3 One machine, two sets of units

THE BANK · A ROOM · PROTOCOL

SCENE 35W

The depth indicator on the bank is driven off the drum shaft. So everything Ubah reads about the cage is really something about the drum. The conversion happens in a gearbox she has never seen.

GUIDE 50W

Each drum quantity on the left has one cage quantity on the right that it fixes. Every conversion in the list is the radius doing one of its jobs. What separates the pairs is which power of the radius each conversion needs. One of the four needs none at all.

ASK 10W

Match each drum quantity to the cage quantity it fixes.

TAKES AS READ

a rope that does not slip covers its own length as the drum turns

VERDICT 88W

A rope that does not slip ties the cage to the drum. The cage's distance, speed and acceleration are the drum's angle, angular velocity and angular acceleration, each multiplied by the radius. That is one relation used three times. It is why the depth indicator can be a gearbox rather than an instrument in the shaft. The coil count is the odd one out. It says how the rope is stacked. That changes the effective radius slightly, and is why a drum is made wide rather than deep.

TAKEAWAY 16W

Radius converts between the drum's angles and the cage's metres. Every rate converts the same way.

4.4 Eleven tonnes, hung from what — Review

ROPE SHOP · A ROOM · CALLBACK · PROTOCOL

SCENE 43W

Kit Nowak, the rope splicer, has the ore skip's rope on the shop board. Four hundred metres of it hang below the sheave, with the skip and its load on the end. The submission puts every one of those forces in one line.

GUIDE 54W

Four forces and four bodies, one each. Ask of every force which body it acts on, rather than where on the plant it happens. A force belongs to a diagram and not to a place. Two of the four act on the same body. Telling those two apart is what the pairing turns on.

ASK 12W

Match each force to the body whose free-body diagram it belongs in.

TAKES AS READ

a rope's tension at a point carries everything hanging below that point

VERDICT 76W

A force enters the equation of whichever body it acts on. The hanging rope pulls down on nothing below it. Each point carries what hangs beneath, so the whole weight is accounted for only in the tension the drum holds. The rope pulls up on the skip, and the ore presses down on it. Both belong to the skip's own line. The steelwork's push belongs to the sheave, which carries both rope ends and accelerates nowhere.

TAKEAWAY 18W

Two forces acting at the same place can belong to different diagrams. Which body each acts on is

END OF THE DAY 18W

A rotating machine needs torque for its own inertia and for the load, and they are separate terms.

Where the work goes

PLAN CARD 123W

Friday. The energy part of the submission is one line long: cage times gravity times depth. Rahimi has asked what happens to the other three quarters. The rope has to be lifted as well. Less of it hangs below the drum with every metre wound on. So the force the drum works against changes the whole way up the shaft. Craig wants the total for her own reason. It is the same energy the brake must take out again if a wind is ever stopped at full speed. Today you take the work done with the rope's weight in it. Then you set the cage's own share beside it, and ask Otieno why he keeps calling twelve hundred metres of steel a spring.

BRIEFING 18W

The energy figures in the submission are the cage's, and the cage is a quarter of the rope.

WHAT YOU DECIDE 17W

Add up the work a wind actually does, with a force that changes all the way up.

5.1 A force that falls all the way up

SHAFT AND BRAKE HOUSE · A ROOM · DERIVE

SCENE 37W

The cage weighs four tonnes loaded and the rope 8.9 kilograms a metre. As the cage rises, less rope hangs below the drum. Rahimi wants the total work for one wind from the inset to the bank.

GUIDE 67W

You build the working a line at a time. The force needed depends on how much rope is still hanging. So it is a function of position rather than a number. Work against it is an accumulation over the 1,240 metres. Write the hanging length in terms of how far the cage has come, then integrate. The wind is taken as steady, so no acceleration term appears.

ASK 15W

How much work does one wind from the inset to the bank do against gravity?

BEHIND THE BUTTON 186W

Why the rope's share is a half rather than a whole. At the bottom of the wind the whole rope is hanging and at the top almost none of it is, and the hanging length falls linearly in between. Integrating a linear force gives the mean value times the distance, which is why the rope contributes as though half of it were being lifted the whole way.

Why the answer is the same by either route. Work against gravity is path-independent. So the same total comes out of asking how much each element of rope was raised. The element at the top of the shaft hardly moves. The one at the bottom is raised the whole depth. That is a different integral with the same answer, which is what makes gravity conservative.

What the number is for. 116 megajoules is what the motor puts in going up and what the brake has to take out coming down. At the shaft's four winds an hour, divided by the cycle time, it is also the plant's electricity bill. That is the other reason Marchetti wants the cycle shorter.

TAKES AS READ

work against a changing force is the accumulation of force over distance

VERDICT 92W

The work is about 116 megajoules, of which the cage accounts for 49 and the rope for 67. The force is not constant. At the bottom of the wind the drum lifts the cage and 1,240 metres of rope. At the top it lifts the cage and nothing. Writing the hanging length as the depth less the distance travelled turns the force into a linear function of position. Integrating it gives the cage's term plus half the rope's. The rope's own weight therefore does more of the work than the load does.

TAKEAWAY 21W

With a force that falls linearly, the work is the same as if the mean force had acted the whole way.

5.2 Forty-nine of a hundred and sixteen

WINDER HOUSE · A PERSON · BALLPARK

SCENE 39W

Marchetti has the submission's energy line on the desk and the new total beside it. She wants the cage's own share computed on its own, so that the difference between the two figures is attributable rather than argued about.

GUIDE 59W

Five quantities are offered and three of them make the answer. The body is the cage and its load, raised the full depth of the shaft. The rope's mass is on the list because it is what the other sixty-seven megajoules are. The speed is there because energy against gravity does not depend on how fast anything was going.

ASK 11W

How much of the wind's work is the cage's own weight?

TAKES AS READ

energy stored against gravity depends only on the change in height · work as a force through a displacement, and as an integral — taken as read

VERDICT 82W

The cage's share is 48.7 megajoules of 116, which is forty-two per cent. Gravity is a conservative force. So this depends only on the mass and the height, not on the route, the speed or the time. That is what lets it be computed as a product rather than as an integral. The rope cannot be treated the same way, because different parts of it are raised by different amounts. That is the whole reason its contribution needed an integral this morning.

TAKEAWAY 17W

Energy stored against gravity depends on the mass and the height. It depends on nothing about the

5.3 Twelve hundred metres of spring

ROPE SHOP · A ROOM · CHOICE

SCENE 34W

Otieno has a length of rope in the shop clamped between two anchors and a dial gauge on it. Pull on it and it stretches; let go and it comes back and goes past.

GUIDE 54W

All four options describe something a long steel rope does. Ask of each whether it would produce a to-and-fro motion of the cage. Then ask what the force would have to be like for that motion to have a definite period. The alternative is a shape that depends on how hard it was pulled.

ASK 13W

What makes the rope behave like a spring rather than like a link?

TAKES AS READ

a force proportional to a displacement always acts back towards where it started

VERDICT 83W

The rope stretches in proportion to the pull, and its pull acts back towards where the cage started. That is what a spring is. It is the condition for an oscillation with a period of its own. A restoring force proportional to displacement gives a to-and-fro motion. That motion's period depends on the stiffness and the mass, not on how far it was pulled. A rigid link would move the cage exactly as the drum moved. Twelve hundred metres of steel does not.

TAKEAWAY 16W

A restoring force proportional to displacement is what makes an oscillation. It gives the motion a

5.4 Three things out of one trace — Review

THE BANK · A ROOM · CALLBACK · SEQUENCE

SCENE 34W

Tom Halloway, the onsetter, has the winder house tachometer roll beside the depth recorder's paper. This pen records speed against time and nothing else, and the inquiry wants the overrun distance out of it.

GUIDE 53W

Four steps, in the order they have to be done. This roll records a rate rather than a position. The quantity the inquiry wants is not on the paper. It has to be accumulated off the roll. Each step needs the one above it to have happened. That is what fixes the order.

ASK 14W

Order the four steps that get the overrun distance out of a speed record.

TAKES AS READ

the area under a speed–time trace is a distance

VERDICT 85W

A record of speed is a rate, so the total is not on the paper. Distance is the area under the trace, not anything read off it. That is the opposite move from the depth recorder. There the totals are on the paper, and the rates are its slopes. Both limits have to be fixed before an area means anything. An area is always between two of them. Only then does subtracting the distance to the landing say how far past it the cage went.

TAKEAWAY 20W

A record of a rate gives up its totals only to an integration. An area is worth nothing until both

END OF THE DAY 21W

Work done against a force that changes with position is an integral, and the changing part can be the larger part.

Ore in a stream

PLAN CARD 134W

Monday. The conveyor drive tripped twice on Friday's night shift. Bart Lindqvist, the tip foreman, has the loads written down. Nothing over four tonnes. Nothing outside what the belt is rated for. What he has not got is the force the belt feels from ore landing on it without a break. That is a different quantity from the weight of the ore sitting on the belt. Rahimi wants the same sum for the arrestors. A mass stopped by crushing steel is the same problem with the sign turned round. Today you find the force in the ore stream. You also settle who feels the push back from it, and go out to the gravity station to see Femi Adebayo, the survey assistant. He will tell you why one reading needs three more around it.

BRIEFING 16W

The conveyor drive trips twice a shift and every load on it is inside its rating.

WHAT YOU DECIDE 16W

Get the force a moving stream makes, and settle who feels the other end of it.

6.1 Three hundred and twenty tonnes an hour

TIP AND CONVEYOR · A ROOM · DERIVE

SCENE 38W

Ore leaves the chute onto the belt at 320 tonnes an hour with no forward speed of its own. The belt runs at 2.5 metres a second. Lindqvist wants the force that takes to keep the belt running.

GUIDE 62W

You build the working a line at a time. Consider the ore that lands in a short interval: it arrives with no forward speed and leaves with the belt's. Something has to have given it that momentum, and dividing the momentum given by the interval is the force. The rate has to be in kilograms a second before any of this works.

ASK 12W

What horizontal force does the ore stream put on the belt drive?

BEHIND THE BUTTON 177W

Why this is not the weight of the ore on the belt. The weight is carried by the idlers and acts downward. The force computed here is horizontal, it exists only while ore is arriving, and it is what the drive has to supply on top of everything else.

Why a stream is not a series of loads. A four-tonne skip tipped into a bin is one event with one impulse. The same ore spread over an hour arrives continuously. What the drive feels is not a series of jolts. It is a steady force equal to the rate times the speed change. Sizing the drive from the loads misses it entirely.

What the vertical part does. The ore also falls about three and a half metres before it lands, arriving at over eight metres a second downward, which the belt and its idlers stop. That is a larger force than the horizontal one. It is carried by the structure rather than by the drive. That is why the drive trips and the frame does not bend.

TAKES AS READ

a force acting for a time changes momentum by their product

VERDICT 86W

The force is about 222 newtons. Every kilogram of ore that lands has to be brought up to belt speed. That needs a momentum of 2.5 kilogram metres a second per kilogram. And 88.9 kilograms land every second. Dividing the momentum given per interval by the interval leaves the rate times the speed change, with the interval cancelling. That is why the answer does not depend on how the interval is chosen. It is a modest force and it is one nobody had included at all.

TAKEAWAY 21W

A mass rate times a speed change is a force, and it is what a drive feels from anything arriving continuously.

6.2 Both ends of one force

SHAFT AND BRAKE HOUSE • A PERSON • CHOICE

SCENE 40W

Rahimi has a crush-tested arrestor from the maker's rig in the corner of the brake house, folded to about half its original height. Whatever it did to the test mass, the test mass did the same thing back to it.

GUIDE 56W

Four accounts of what happened between the falling mass and the arrestor on the rig. Ask of each which body each force acts on. A force and its partner never appear in the same free-body diagram. The distinction that decides this card is between two forces on one body and one force seen from both ends.

ASK 17W

What is the relation between the force on the falling mass and the force on the arrestor?

TAKES AS READ

forces come in pairs acting on different bodies

VERDICT 81W

They are one interaction seen from both ends. Equal in size, opposite in direction, and acting on different bodies. That is why they do not cancel. A cancellation would need both forces on the same body. The mass's diagram has the arrestor pushing up on it and gravity pulling down. The arrestor's diagram has the mass pushing down on it and the rig holding it up. Each diagram is solved on its own, and the pair ties the two solutions together.

TAKEAWAY 16W

An interaction is one thing seen twice, and its two halves belong to two different diagrams.

6.3 Four readings for one number

GRAVITY STATION · A ROOM · CASEBOOK

SCENE 37W

Adebayo has the field book open at the bench station. On the page: a reading, a time, an elevation. Then two more readings of the same reference station, taken hours apart at either end of the day.

GUIDE 49W

Each entry in the field book on the left is there to answer one question on the right. What separates them is which of them is about the ground and which is about the instrument. The one about the instrument is the reason the same station is read twice.

ASK 13W

Match each entry in the field book to what it is there for.

TAKES AS READ

an instrument's own behaviour can change over the hours a survey takes

VERDICT 79W

A gravimeter is a weighted spring, and a spring creeps. So the same station read twice, hours apart, gives the drift over that interval, and the times of the readings in between are what let it be apportioned. Elevation matters because gravity falls with height at about 0.3 milligals a metre, which is far larger than the rock signal being looked for. What is left after both corrections is the measurement, and it means nothing at all without them.

TAKEAWAY 19W

A reading becomes a measurement only once the instrument's own drift and the station's own height are accounted for.

6.4 Where twelve seconds come from — Review

WINDER HOUSE · A ROOM · CALLBACK · BALLPARK

SCENE 44W

Ruth Anand, the winder driver, has the cycle sheet up in the winder house. Eleven hundred and fifty metres of the wind are run at full speed. The ramps are at one end and the crawl at the other. Marchetti wants that part timed.

GUIDE 55W

Four quantities are offered and two of them belong in the sum. The section in question is the one where the winder is neither speeding up nor slowing down. So nothing has to be integrated, and no ramp figure enters it. The other two numbers describe parts of the cycle that are not this one.

ASK 10W

How long does the full-speed section of one wind last?

TAKES AS READ

a wind is made of a ramp up, a run at full speed, a ramp down and a crawl

VERDICT 79W

About a hundred and sixty seconds, and none of them are anybody's to give away. The full-speed run is a distance divided by a speed. The distance is the shaft less the ramps at each end. The speed is what the licence allows. Neither moves because a ramp is made steeper. That is why the twelve seconds have to come out of the ramps and the crawl. Those are the only parts of the cycle a winding engineer chooses.

TAKEAWAY 21W

The longest part of a winding cycle is set by the depth and the permitted speed. It is not a place

END OF THE DAY 22W

A stream of mass arriving at a rate makes a steady force, and no single load in it accounts for that force.

A kilometre and a quarter of rock

PLAN CARD 142W

Tuesday. Dr. Lena Sokol, the geophysicist, has the shaft for two hours on Thursday. That is the third time in four years anyone has been given it. She has one reading at the bench station and one taken at the inset. The difference between them comes out the wrong way round. Gravity is larger at the bottom of the shaft. The uniform-sphere result in every textbook says it should be smaller. Nobody at this mine has ever looked at that. Today you work out what gravity does inside a sphere. From the disagreement you can get the density of the rock above the inset. You also settle with Kit Nowak, the rope splicer, what six months of shortening has done to the rope. A working shaft is the only place in the country that can take a gravity meter down a kilometre.

BRIEFING 16W

Sokol has two readings a kilometre and a quarter apart, and they disagree with the textbook.

WHAT YOU DECIDE 15W

Work out what gravity should do down the shaft, and what the measurement says instead.

7.1 The shells above you

GRAVITY STATION · A ROOM · DERIVE

SCENE 37W

Sokol has the shell theorem written on the hut wall and two readings on the bench: one at the surface, one 1,240 metres down. She wants the prediction for a uniform earth first, and then the disagreement.

GUIDE 56W

You build the working a line at a time. The shell theorem says a spherical shell of matter pulls nothing that is inside it, so at a radius part-way down only the sphere below counts. Write that sphere's mass in terms of the density, put it into the inverse-square law, and see what the r's do.

ASK 15W

What should gravity do over the 1,240 metres of this shaft, for a uniform earth?

BEHIND THE BUTTON 191W

Why the shells above contribute nothing. Every part of a shell pulls you towards it. For a point inside the shell those pulls cancel exactly. The nearer piece of shell is smaller and the further piece larger, in exactly compensating proportion. It is a result about geometry rather than about gravity, and it is what makes the inside of a planet tractable at all.

Why the inverse square becomes a straight line. Only the mass below counts, and that mass falls as the cube of the radius while the inverse-square law divides by the square. A cube over a square is a first power. So gravity inside a uniform sphere is proportional to how far out you are. It reaches zero at the centre.

What the disagreement is worth. The prediction for a uniform earth is a fall of 191 milligals over this shaft; the measurement is a rise of 105. The earth is not uniform. Most of its mass is far in. Locally, what the difference constrains is the density of the rock between the two stations. That is the number the exploration people have been assuming rather than measuring.

TAKES AS READ

a spherical shell of matter exerts no net pull on anything inside it · newton's law of gravitation — taken as read

VERDICT 89W

A uniform earth would give a fall of about 191 milligals. The shell theorem removes everything above the measurement point. So what pulls is a sphere whose mass goes as the cube of the radius. Dividing that by the square of the radius leaves a straight line through the centre. The measurement is a rise of 105 milligals instead. That is not an error. The real earth has most of its mass concentrated far inside. The local rock is less dense than the average. The disagreement is the measurement.

TAKEAWAY 21W

Inside a uniform sphere gravity grows in proportion to the distance from the centre, because only the mass below you counts.

7.2 Two thousand six hundred and seventy

GRAVITY STATION · A PERSON · BALLPARK

SCENE 49W

Sokol has the two gradients on the bench. One is what gravity would do down a shaft with nothing in the walls. The other is what it actually does. The difference is the rock between the stations, counted twice, because rock that was above you stops being above you.

GUIDE 67W

Five quantities are offered and three of them make the answer. The free-air figure is what gravity does with height alone. The measured figure is what this shaft gives. The coefficient turns a density into a gradient. Two are not needed. The depth cancels out of a gradient, and the mean density of the whole earth belongs to the uniform sphere this shaft has just disagreed with.

ASK 13W

What density does the measured gradient give for the rock above the inset?

TAKES AS READ

rock that was above the instrument stops being above it as the instrument goes down · the field inside a uniform sphere, and the shell theorem — taken as read

VERDICT 91W

The rock comes out at about 2.67 grams a cubic centimetre, which is granite. Going down a shaft does two things at once. It takes you closer to the centre, which is the free-air term. And it puts rock above you that was below you. That removes that rock's upward pull twice over. The measured gradient is the difference between them, so the density is what is left when one is taken from the other. It says nothing whatever about what lies below the inset — that is left entirely free.

TAKEAWAY 21W

A gravity gradient down a shaft constrains the density of the rock passed through, and constrains nothing about what is deeper.

7.3 Forty metres shorter

ROPE SHOP · A ROOM · CHOICE

SCENE 39W

Nowak has the rope record: the cappel has been re-capped four times in two years and each re-cap takes ten metres off. The rope is forty metres shorter than the file says and made of exactly the same steel.

GUIDE 60W

Four claims about what shortening a rope changes. Ask of each whether it follows from the rope having less length, or from it having less material. Those are different things here. The steel is unchanged and only its length has gone. One of the four is a property that depends on the length and is nowhere in the winder's paperwork.

ASK 11W

What has forty metres of shortening changed that nobody has recorded?

TAKES AS READ

a shorter piece of the same material stretches less under the same pull

VERDICT 94W

It has made the rope stiffer, and therefore made the cage bounce on it faster. Stiffness for the same steel goes inversely with length. A rope forty metres shorter out of twelve hundred and eighty is about three per cent stiffer. So the period of the cage on it is about one and a half per cent shorter. That is small, and it matters. The whole question on day nine is whether that period lands on a ramp time. A period nobody has recomputed since the file was written is a period nobody knows.

TAKEAWAY 19W

A property that depends on a length changes whenever the length does, whether or not anybody writes it down.

END OF THE DAY 21W

Gravity inside a body depends only on the mass below you, which is why a shaft measures what a surface cannot.

The worst instant

BEFORE THE DAY — FOLLOW

The rope inspection route, walked by the man who splices it

Kit Nowak, the rope splicer, walks the rope from the drum to the sheave to the cappel on the same route every week, and the route is not written down anywhere. Stay with him along it. Where he stops is where the rope is watched, and the forty metres that have been cut off it in two years were all cut at one of those stops.

PLAN CARD 130W

Wednesday. The electrical engineer has asked a question nobody in the winder house can answer. At which instant of a wind is the motor working hardest? Marchetti has the torque, which is largest at the bottom. She has the speed, which is largest in the middle. A motor is rated by the two of them multiplied together. Craig wants the same figure from her own side of the shaft. Whatever the motor puts in on the way up is what the pads take out on the way down. Today you find the power at the hardest instant of the wind. You then put four instants of the wind in order by what each one asks for. Craig will settle with you what the brake is really being asked to do.

BRIEFING 16W

The motor is rated in kilowatts and the submission has never said where the peak is.

WHAT YOU DECIDE 15W

Find where in the cycle the motor works hardest, and put a number on it.

8.1 A megawatt and a bit

THE BANK · A ROOM · DERIVE

SCENE 36W

Just past the end of the ramp the cage is at 7.2 metres a second with 1,182 metres of rope still hanging below the drum. Ubah wants the power the motor is supplying at that instant.

GUIDE 69W

You build the working a line at a time. Power is the rate at which work is being done. For a force along the motion, that is the force times the speed at that instant. So establish two things. What is the tension where the cage happens to be, and how fast is it going there? The wind is steady at this instant, so there is no acceleration term.

ASK 13W

What power is the motor supplying just after the end of the ramp?

BEHIND THE BUTTON 183W

Why just after the ramp. The tension falls all the way up, as rope comes off the hanging side. The speed rises to its maximum during the ramp and then holds. So the product is largest at the moment the speed has just arrived and the rope has hardly begun to shorten. That is neither of the instants anybody quotes.

Why the acceleration term is absent here. During the ramp the tension is higher, by the hanging mass times 1.2 metres a second squared. But the speed is lower. The two do not peak together. At the end of the ramp the acceleration has tapered to nothing by design. That is what makes this instant the worst one, and the arithmetic the simplest.

What the answer settles. The motor is rated at 1.2 megawatts continuous, so 1.03 is inside it. The faster cycle raises the peak speed rather than the torque. That is where Marchetti's proposal actually presses. Power scales with the speed. So the twelve seconds she wants cost about nine per cent more peak power, and no more torque at all.

TAKES AS READ

power is the rate at which a force does work

VERDICT 84W

The motor is supplying about 1.03 megawatts. Power is the rate of doing work. For a force acting along the motion, it is the force times the speed at that instant. It is not anything averaged. The tension has to be evaluated where the cage actually is: 1,182 metres of rope still hanging, not the full 1,240. The speed is the one the ramp has just delivered. The peak is here rather than at the bottom, because the two factors do not peak together.

TAKEAWAY 22W

The peak power of a wind is at the top of the ramp. The speed has arrived there, and the rope has

8.2 Four instants, one rating

WINDER HOUSE · A PERSON · SEQUENCE

SCENE 30W

Marchetti has four instants of an upward wind written on the board and the motor's rating beside them. The electrical engineer wants them in order before he will sign anything.

GUIDE 59W

Four instants, and the rail is not a clock. All four happen in one wind. You are ranking them, not sequencing them. What decides the order is the product of a force and a speed at each instant. So an instant with a large force and no speed can rank below one with a modest force and full speed.

ASK 13W

Order the four instants by the power the motor is supplying, largest first.

TAKES AS READ

a power is a force times a speed and has the units of both

VERDICT 91W

Power is a product, so the order is not the order of either factor. Just past the ramp both are large: 142 kilonewtons at 7.2 metres a second, about 1.03 megawatts. Half way up the speed is the same and the tension has fallen to 93, giving 0.67. The first instant off the inset has the largest tension and a speed near zero. Its power is near zero too. That is why a winder can start a load it could not accelerate. The crawl at the bank is small in both.

TAKEAWAY 16W

Ranking instants by power means multiplying at each one, because neither factor alone decides the order.

8.3 A force or an energy

SHAFT AND BRAKE HOUSE · A ROOM · CHOICE

SCENE 34W

Craig has the pad wear records for two years. The pads last well on ordinary winds and the set after March had to be replaced. Nothing about the force the brake applies was different.

GUIDE 62W

All four options are ways of saying what a brake does. Ask of each whether it is a force, an energy or a rate. Those are three different quantities. The pads only wear out from one of them. The distinction that decides it is what has to be taken out of the cage rather than what has to be applied to it.

ASK 10W

What decides how hard a stop is on the pads?

TAKES AS READ

work done against a body's motion is a change in its kinetic energy

VERDICT 99W

It is the energy: whatever the cage has as motion and as height has to be turned into heat in the pads. The work–energy theorem says the net work on the cage is its change in kinetic energy. So the same energy can be taken out with a large force over a short distance, or a small one over a long one. The pads feel the total either way. That is why the March stop destroyed a set of pads while applying the same force as every other stop. It went on for longer. There was more to absorb.

TAKEAWAY 24W

A brake is sized by the energy it has to absorb, and by the force only in so far as that decides the distance.

END OF THE DAY 18W

Power is a product, so its peak is at neither of the places where its two factors peak.

What the rope does on its own

PLAN CARD 145W

Thursday. Otieno has the shaft's rope data out on the shop bench. There is the area of steel and the modulus the maker quotes. There is the length the rope is today, not the length in the file. The modulus is how much the steel stretches under a pull. Those three give a stiffness. The stiffness gives a period. A period is how long the cage takes to bounce once on the end of the rope. Nothing in the submission has a period in it. Nothing in the winder handbook has one either. Today you take the rope's stiffness and its period. You then settle what a bounce stores while it is happening, and ask Ubah what the crawl at the bank is really for. The rope stretches more than a metre under an ordinary load. Nobody has put that number beside the overwind margin.

BRIEFING 15W

Otieno has asked for one number for two years and it has never been computed.

WHAT YOU DECIDE 16W

Get the stiffness of twelve hundred metres of steel and the period it gives the cage.

9.1 A hundred and ten kilonewtons a metre

ROPE SHOP · A ROOM · DERIVE

SCENE 43W

The rope has 1,140 square millimetres of steel in it, and a modulus of 120 gigapascals as a locked coil. It is 1,240 metres long with the cage at the inset. Otieno wants the stiffness and the period the cage has on it.

GUIDE 68W

You build the working a line at a time. Stiffness is how much pull it takes to stretch the rope by a metre. It comes from the material, the area and the length. Then the cage on the end of it is a mass on a spring. The period follows from the stiffness and the mass. Watch where the length goes: a longer rope is a softer spring.

ASK 16W

What period does the cage have bouncing on the rope, with the cage at the inset?

BEHIND THE BUTTON 177W

Why length divides. Doubling the length of a rope gives twice as much material to stretch under the same pull. So it stretches twice as far, and the stiffness halves. That is why deep shafts have soft ropes. A laboratory sample tells you nothing about the shaft unless the length is carried through.

Why the period does not depend on the load's weight in the way people expect. It depends on the mass hanging and on the stiffness, not on how far the rope is already stretched. So a cage with four tonnes in it and the same cage empty have different periods. The same cage at the inset and at the bank have very different ones. The rope is ten times softer at the bottom.

What the static stretch is. Under an ordinary 147 kilonewtons this rope is stretched by about 1.3 metres. That is not a fault. It is a metre and a third of stored energy sitting in the rope on every wind. It is the reason the next day's question exists at all.

TAKES AS READ

a rope's stretch is proportional to the pull on it

VERDICT 86W

The period is about 1.2 seconds. The rope's stiffness is its modulus times its steel area divided by its length. That comes out at 110 kilonewtons a metre — soft, because the rope is very long. A four-tonne cage on that spring has an angular frequency of 5.25 radians a second. The period is 2π divided by it. Neither the stiffness nor the period depends on how far the rope is already stretched. That makes them properties of the installation rather than of a particular wind.

TAKEAWAY 19W

Stiffness is the material times the area over the length. The period the cage has on it follows from

9.2 A metre and a third of stretch

SHAFT AND BRAKE HOUSE · A PERSON · CHOICE

SCENE 34W

Rahimi has the stretch figure on the board. It is a metre and a third under an ordinary load. It was measured against the guides and confirmed against the depth indicator on two winds.

GUIDE 53W

Four accounts of what that stretch means. Ask of each whether it describes something the rope is doing or something the rope is holding. Then ask what would happen to it if whatever is pulling let go. One of the four is the reason a cage does not stop when the drum does.

ASK 15W

What is the rope holding while it is stretched by a metre and a third?

TAKES AS READ

a stretched spring holds energy that can be recovered

VERDICT 84W

It is holding energy, about a hundred kilojoules of it, and it will give that energy back to whatever is on the end of it. That is what a spring does: the work done stretching it is stored as the square of the stretch, and released as the stretch comes out. So a cage on a stretched rope is not a load on a link. It is a mass on a charged spring. Stopping the top of that spring does not stop the mass.

TAKEAWAY 23W

A stretched rope holds energy proportional to the square of its stretch, and it gives it back to whatever is on the end.

9.3 Three seconds of nothing much

THE BANK · A ROOM · CHOICE

SCENE 31W

Ubah's landing has a crawl of about three seconds at the end of every wind, and Marchetti's proposal takes two of them out. Ubah has watched some four thousand winds arrive.

GUIDE 39W

All four options are things the crawl does. Ask of each whether it would still be needed on a rigid link, rather than on twelve hundred metres of steel. That is what separates a working margin from a habit.

ASK 11W

What is the crawl at the end of a wind doing?

TAKES AS READ

an oscillation loses amplitude over time to whatever damps it

VERDICT 82W

It gives whatever the cage is doing on the rope time to die away before the stop. A cage arriving at speed is bouncing, at a bit over a second a cycle. The damping in the rope and the guides takes several cycles to quieten it. Three seconds is two or three of those cycles. On a rigid link there would be nothing to wait for. That is why the crawl looks like caution to anybody reading the profile, and is not.

TAKEAWAY 17W

A margin that reads as caution is often a quantity somebody once computed and nobody wrote down.

END OF THE DAY 17W

A hoisting rope is a spring with a period, and neither appears anywhere in the plant's paperwork.

A metre and six tenths

PLAN CARD 138W

Friday. Everything the March inquiry did not have is now on one board. The rope's stiffness. The cage's period. The energy stored in a stretch. Anand put the emergency brake on with the cage going down, forty metres above the inset, at eight and a half metres a second. The drum stopped in under a second. The cage's own position recorder shows it moving a metre and six tenths after that. It came to rest below the landing. Craig's pad measurement was right, and it was taken cold on a bench. Today you work out how far a cage travels after its drum has stopped. You then decide which account of March fits every record, and settle with Anand what a brake actually stops. This is the day Otieno turns out to have been right for two years.

BRIEFING 17W

The inquiry blamed the pads. The pads were fine, and the cage still went past the landing.

WHAT YOU DECIDE 12W

Account for the March overwind with the numbers now on the board.

10.1 What the brake does not stop

SHAFT AND BRAKE HOUSE · A ROOM · DERIVE

SCENE 36W

The drum was stopped from 8.4 metres a second in under a second, with the descending cage forty metres above the inset on twelve hundred metres of rope. Rahimi wants how much further the cage went.

GUIDE 62W

You build the working a line at a time. Take the drum as stopped and the cage as a mass on a spring anchored at the drum. All the cage's motion has to be turned into stretch, so the energy it arrives with is the energy the rope ends up holding. Solve for the stretch, and note what the answer depends on.

ASK 11W

How far past the drum's stopping point does the cage travel?

BEHIND THE BUTTON 186W

Why the answer is the speed over the angular frequency. Set the cage's kinetic energy equal to the rope's stored energy and solve for the stretch. It comes out as the speed times the root of mass over stiffness. That is the speed divided by the angular frequency computed yesterday. So it is not an arbitrary quantity: it is one period's worth of travel, near enough, at the speed the cage was making.

Why no brake could have prevented it. The brake acts on the drum. Whatever it does to the drum, the cage is a separate body attached through a spring, and its own momentum has to be taken out through that spring. A better pad stops the drum sooner and makes this worse rather than better, because the drum stopping is what starts the overshoot.

What the inquiry could not have seen. Craig's coefficient was correct. The retardation computed from it was correct for the drum. Every number in the finding was right. The model behind them had the cage and the drum as one rigid body. Twelve hundred metres of steel is not.

TAKES AS READ

all of a body's kinetic energy has to be accounted for by work done on it

VERDICT 91W

The cage travels about 1.6 metres further, which is what was measured. Once the drum is stopped the cage is a mass on a spring, and all of its kinetic energy has to go into stretching the rope. Equating the two and solving for the stretch gives the speed divided by the angular frequency — 8.4 over 5.25. Nothing about the pads appears in that expression. The overwind was a property of the installation and the speed at which the brake was applied, and both were within the rules as written.

TAKEAWAY 19W

A cage on a soft rope overshoots its stopped drum by its speed divided by its own angular frequency.

10.2 Four records, one account

ROPE SHOP · A PERSON · DIAGNOSIS

SCENE 35W

Everything from March is on the shop bench. There is the pad certificate, and the drum's own record of when it stopped. There is the cage's position tape, and the pads that were replaced afterwards.

GUIDE 63W

Look at every record before choosing, including the two that are not alarming. The pad certificate is a good measurement and the drum record shows a stop made on time. What separates the accounts is whether they explain a cage that moved after the drum had stopped. The second test is pads worn out by a stop they were strong enough to make.

ASK 6W

Which account fits all four records?

TAKES AS READ

energy taken out of a moving body has to appear somewhere as heat or as work

VERDICT 101W

The drum record and the pad certificate are both in order. That rules out the two accounts that need one of them to be wrong. What is left has to explain a cage that travelled after the drum had stopped, and pads worn through by a stop they were strong enough to make. A cage on a rope of 110 kilonewtons a metre overshoots a stopped drum by 1.6 metres at that speed. The pads then take out the cage's energy as well as the drum's. That energy arrives back through the rope over the following seconds, not during the stop.

TAKEAWAY 20W

Pads wear from the energy they absorb. A stop that goes on longer absorbs more of it at the same

10.3 What the brake has hold of

WINDER HOUSE · A ROOM · CHOICE

SCENE 30W

Anand has been saying for eight months that the machine stopped and the cage did not. There is no instrument in the winder house that would show her either way.

GUIDE 61W

All four options are bodies the brake could be said to stop. Ask of each what the brake is physically in contact with. Then ask what has to happen for a body not in contact with it to slow down. The depth indicator is the detail that decides it. What it reads is one of the four and not the others.

ASK 7W

What does the brake have hold of?

TAKES AS READ

a brake acts on whatever it is in contact with

VERDICT 84W

The pads clamp the drum flanges, so the brake stops the drum. Everything else slows down only through the rope, and the rope can stretch. So while the drum is being stopped, the relation between drum position and cage position quietly stops holding. The depth indicator is driven off the drum shaft. It reads where the cage would be if the rope were rigid. In March it read the cage stopping at the mark. The cage was a metre and six tenths past it.

TAKEAWAY 19W

A no-slip relation ties a drum to a rope. It stops being a relation the moment the rope stretches.

END OF THE DAY 17W

A brake stops the drum; what stops the cage is the rope, and the rope stretches first.

Everything the pads have to take

PLAN CARD 134W

Monday, and two working days are left before the inspector. Craig has the pads' rating in front of her. The maker says six hundred kilojoules a stop. Nobody has a figure for what one stop actually puts into them. Three things are moving when the brake goes on. The cage. The rope hanging with it. And eighteen tonnes of drum, which is the one nobody counts. Marchetti has agreed to one test wind on the empty cage this afternoon. Today you total the energy an emergency stop delivers. You then predict what the test wind will do, and go and measure it. Lindqvist will show you what a skip tipping into a bin has in common with all of it. Whatever the pads cannot take in is what the shaft finds out about later.

BRIEFING 16W

The pads are rated per stop and nobody has added up what a stop actually contains.

WHAT YOU DECIDE 17W

Add up what an emergency stop puts into the brake, and test one prediction against a wind.

11.1 Five hundred and sixty kilojoules

WINDER HOUSE · A ROOM · DERIVE

SCENE 43W

An emergency stop, with the cage going up and half way. It is making 8.4 metres a second. There are 620 metres of rope hanging. The drum is turning at 4.0 radians a second, and everything comes to rest in about 3.8 metres.

GUIDE 57W

You build the working a line at a time. Three stores have to be emptied. They are the drum's rotation, the motion of everything travelling, and whatever gravity does over the stopping distance. The third one has a sign that depends on which way the cage is going. It is the term the whole answer turns on.

ASK 12W

How much energy does an upward emergency stop put into the pads?

BEHIND THE BUTTON 167W

Why the drum is the largest single store. Eighteen tonnes at two metres' radius has an inertia of 72,200 kilogram metres squared. At 4.0 radians a second that is 578 kilojoules. It is more than the cage and its rope together. It is invisible on the plant drawings because nothing about it moves in a straight line.

Why the direction of travel decides the answer. Going up, gravity is taking energy out of the cage while the brake is, so the pads absorb less: 558 kilojoules. Going down, gravity is putting energy in, and the same stop asks the pads for 1.27 megajoules. The stop that destroyed a set of pads in March was a downward one.

What the rating means. Six hundred kilojoules a stop is the maker's figure for the pads fitted. An upward emergency stop is inside it, and a downward one is twice it. That is not a fault in the pads. It is a fact nobody at this shaft had written down.

TAKES AS READ

a turning body holds energy that has to be taken out of it to stop it

VERDICT 88W

The pads take about 558 kilojoules, against a rating of 600. Three stores are being emptied, and the drum is the biggest of them. A rotational energy is a half times the moment of inertia times the square of the angular velocity. This drum's inertia is large. Going up, gravity helps: the cage rises 3.8 metres while stopping. So 355 kilojoules leaves the system as height, and the pads never see it. Going down the same term arrives with the opposite sign, and the total is 1.27 megajoules.

TAKEAWAY 21W

The energy a stop must absorb is the rotation plus the motion plus or minus what gravity does over the distance.

11.2 Nine tenths of a second

SHAFT AND BRAKE HOUSE · A ROOM · VERIFY

SCENE 31W

Marchetti has released the empty cage for one test wind this afternoon. Craig can apply the emergency brake from 8.4 metres a second and the drum recorder will trace what happens.

GUIDE 66W

Lock a prediction for how long the drum takes to stop before the brake is applied. Nothing about it can be changed afterwards, which is the point of making it first. Then apply the brake. Then spend a test wind on reading the recorder. A stop that is confirmed to have happened and never measured establishes that the brake works. It establishes nothing about how fast.

ASK 15W

Predict how long the drum takes to stop, apply the brake, and report what happened.

TAKES AS READ

a retardation follows from the braking torque and the inertia it acts on

VERDICT 91W

The stop takes about nine tenths of a second. It follows from the braking torque acting on the drum's inertia and on the translating mass through the rope. So a prediction is a test of the model rather than a guess. That is exactly why it has to be locked first. With the trace in hand, a number can be chosen that fits and the prediction will appear to have been right. The commoner failure is the other one: applying the brake, watching everything stop, and treating that as the result.

TAKEAWAY 20W

A prediction locked before an intervention is a measurement of the model; the same number chosen afterwards is a description.

11.3 Four tonnes into a bin

TIP AND CONVEYOR · A PERSON · CHOICE

SCENE 35W

Lindqvist has the bin's mounting bolts on the bench, two of them sheared. A skip tips four tonnes into that bin twenty times a shift and the ore arrives with a speed of its own.

GUIDE 50W

All four options are ways of accounting for what the bolts feel. Ask of each what body or bodies have been taken together, because a momentum that leaves one body has to arrive at another. The one that works is the one where nothing outside the chosen bodies is needed.

ASK 11W

What has to happen to the momentum the ore arrives with?

TAKES AS READ

momentum lost by one body is gained by another unless something outside acts

VERDICT 93W

It has to be taken by the bin, and through the bin by its bolts. Ore and bin together are a system with nothing outside it but gravity and the bolts. So the momentum the ore loses appears as an impulse on the bolts. It is spread over however long the ore takes to stop. Twenty times a shift, at four tonnes arriving at seven metres a second, that is a real load. It was never in the bolt calculation. That calculation was done for the weight of a full bin standing still.

TAKEAWAY 19W

Taking two bodies together removes the force between them from the accounting and leaves only what comes from outside.

END OF THE DAY 20W

A stop has to absorb the rotation as well as the motion, and the rotation can be the larger half.

What gets signed

PLAN CARD 125W

Tuesday, and the inspector arrives in the morning. The fortnight has settled two things. Marchetti was right about the machine. Otieno was right about the rope. The motor has the torque and the power it needs. What binds the profile is the energy in a downward emergency stop, and how far a cage travels after its drum has stopped. A revised profile has been drafted. It keeps most of the saving by capping the speed at which the brake may be applied. Today you integrate that profile and say what the cycle time really comes to. You then spend what is left of the maintenance budget, and settle with Rahimi what goes in the submission. Forty-one men a shift. Four winds an hour. One signature.

BRIEFING 16W

The inspector is here tomorrow. Two of the twelve seconds are not available and seven are.

WHAT YOU DECIDE 15W

Integrate the revised profile, spend what is left, and say what the licence rests on.

12.1 Seven seconds, not twelve

THE BANK · A ROOM · DERIVE

SCENE 45W

The revised profile keeps the 12-second ramp. It keeps the full speed of 7.2 metres a second. It brings the cage down to 4.5 metres a second before the brake may be used. It cuts the crawl from three seconds to one and a half.

GUIDE 62W

You build the working a line at a time. The cycle is a sum of pieces. Each piece is either a distance divided by a speed, or an integral of a profile. Work out the shaft the ramps use, take what is left for the full-speed run, and add the crawl. Then compare with the present cycle rather than with the proposal.

ASK 10W

What does the revised profile take off the cycle time?

BEHIND THE BUTTON 172W

Why the retardation cap is what costs the seconds. Capping the speed at which the brake may be applied changes the retardation ramp. The ramp has to bring the cage down to 4.5 metres a second before the mark. That takes shaft, and therefore time. It is five of the twelve seconds Marchetti asked for, and it is the price of an overshoot inside a metre.

Why the crawl can be halved. The crawl exists to let the bounce die away, and the bounce after a retardation ramp is far smaller than the bounce after a brake application. With the brake capped, one and a half seconds is two periods, which is enough.

What the submission now says. Seven seconds a cycle, four winds an hour, eight hours a shift: about three and a half minutes a shift, which is a wind and a half. It is less than Marchetti asked for and it is a real saving, and it rests on a limit nobody would have written down a fortnight ago.

TAKES AS READ

a cycle time is the sum of the times of the parts of the journey

VERDICT 89W

The revised profile saves about seven seconds, not twelve. The ramp up is unchanged at 12 seconds for 57.6 metres. The retardation now has to reach 4.5 metres a second rather than nothing before the brake may be used. That lengthens the ramp, and eats five and a half of the seconds the proposal claimed. The crawl gives one and a half back. What is left is a saving that survives the two constraints the fortnight found. That is the only kind worth putting in front of an inspector.

TAKEAWAY 21W

A cycle time is only as good as the integral behind it. A saving that ignores a limit is not a

12.2 A hundred points before the inspection

WINDER HOUSE · A PERSON · SCIENCETANK

SCENE 33W

Marchetti has a hundred points of maintenance money left in the year and four things it could go on. The inspector reads whatever this fortnight produced, and the licence runs for three years.

GUIDE 75W

What the plant holds already. A pad coefficient measured cold on a bench and quoted for winds. A rope stiffness computed from the maker's modulus and never measured on this rope. A cage position tape at the inset, the only instrument that saw March, and nothing like it at the bank. A drum inertia now taken off the drum's own dimensions. The arrestors above the bank have never been used and have never been tested.

ASK 7W

Spread a hundred points across the four.

TAKES AS READ

money spent on a measurement is worth what the measurement would change

VERDICT 88W

The position recorder is the instrument that would have seen March. There is one end of the shaft where the same effect would be an overwind, not a step down into a gap. Measuring the stiffness turns the number every overshoot rests on into a measured one. The hot friction rig is worth something because Craig's figure has been used where it does not apply. That will happen again. The arrestor test is real work on equipment that is inside its margin and has never been called on.

TAKEAWAY 18W

Evidence is worth what it would change about a decision, not what it would add to a record.

12.3 What the licence rests on

SHAFT AND BRAKE HOUSE · A ROOM · CASEBOOK

SCENE 29W

Rahimi has four findings on the bench and one submission to write. The inspector will read it in the morning and the licence runs for three years after that.

GUIDE 62W

Each finding on the left has one consequence on the right that the submission has to state. What separates them is which of the four is a limit on the machine. Which is a limit on the rope. And which is a limit on how a stop may be made. The licence is a set of limits, not a set of results.

ASK 11W

Match each finding to the limit it puts in the submission.

TAKES AS READ

a limit is only useful if it names the quantity it limits

VERDICT 106W

Four findings and four different kinds of consequence. The motor's margin needs no limit at all, which is the finding that lets the cycle come down. The overshoot is a property of the rope and the speed, so the limit belongs on the speed. The pad energy is about how a stop is made rather than how fast the cage goes, so it limits the stop. And the rope's stiffness changes on a known schedule, so what it needs is not a limit but a standing instruction to recompute. A licence that carried only the first of these would be the submission of a fortnight ago.

TAKEAWAY 22W

A submission is a list of limits, and each one has to name the quantity it constrains and the reason it exists.

END OF THE DAY 24W

A revised profile is a decision about which numbers the plant is allowed to reach, and it has to be integrated to be believed.

THE CLOSING CARD

The licence was granted on a cycle seven seconds shorter, with two limits in it that nobody would have written down a fortnight earlier: a cap on the speed at which the brake may be applied, and a prohibition on emergency stops on the descending wind except in an emergency. Marchetti was right about the machine — the motor reaches 81 per cent of its torque rating and 86 per cent of its power rating and the faster cycle uses the margin it has. Otieno was right about the rope. The March stop is in the file now with an explanation attached: a cage on 110 kilonewtons a metre travels its own speed divided by 5.25 after the drum has stopped, which was a metre and six tenths, and no pad ever made would have changed it.

What it cost: five of the twelve seconds, a set of pads, and the maker's man going home with a set he had driven three hours to fit. What is unfinished: the rope's stiffness is still the maker's modulus rather than a measurement of this rope; there is a cage position recorder at the inset and none at the bank, which is the end where the same effect would be an overwind; and Craig's friction figure is still measured cold on a bench because nothing here can measure it hot.

You differentiated a curve somebody had submitted as a shape, and took the drum's inertia off the drum instead of out of an eighteen-year-old file. You found the tension at the end of the rope where it is worst, and then you found the thing the inquiry could not: that a brake stops the drum and the cage is on a spring. A driver who had been cleared without being believed for eight months has an explanation with her name nowhere near the cause of it. That was your fortnight.

Card text only — no options, boards or answers. 8,543 words of card prose across 40 stops. Generated from themes/overwind by tools/make-cardbook.mjs.